

Gavindya Jayawardena

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PROFILE

Data Engineer and Analyst with 8+ years of experience building robust data pipelines, real-time streaming systems, and NLP-driven workflows across diverse data sources. Skilled in translating complex requirements into actionable data solutions and insights. Published researcher and graduate-level educator with a proven ability to bridge engineering, analytics, and cross-functional teams.

SKILLS

Programming	Python, R, C#, Java, JavaScript, HTML/CSS
Data Engineering	End-to-end pipeline design, real-time & streaming data systems, structured & unstructured data processing, NLP-driven document extraction, signal processing, feature engineering, WebSocket ingestion
Databases	SQL, NoSQL (MongoDB, Cassandra)
Cloud & DevOps	Docker, Git, REST APIs, Flask, Jupyter
ML & Analytics	scikit-learn, PyTorch, TensorFlow, Keras, HuggingFace, NumPy, Pandas, SciPy, Matplotlib, XGBoost, Random Forest, GCN, GROBID, OpenCV, YOLO, Mask R-CNN, ANOVA, time series analysis, statsmodels
Life Sciences	Biosensory, physiological, eye-tracking, pupillometry, EEG, and behavioral data; clinical data analysis; IRB-compliant data collection
Research Methods	Controlled experiment design, IRB protocols, user interviews, surveys, Qualtrics, mixed-effects models, ANOVA
Agile & PM	Iterative research delivery, cross-functional team coordination

EDUCATION

Old Dominion University , Norfolk, VA — <i>Ph.D., Computer Science</i> GPA: 4.0/4.0 Dissertation: “RAEMAP: Real-Time Advanced Eye Movements Analysis Pipeline” <i>Lee Entsminger Outstanding Dissertation Award (2024-2025); CS Outstanding Researcher Award (2019-2020)</i>	<i>Jan 2019 – Aug 2024</i>
University of Moratuwa , Sri Lanka — <i>B.Sc., Computer Science and Engineering</i> GPA: 3.85/4.20 First Class Honors	<i>Jan 2015 – Dec 2018</i>

RESEARCH & WORK EXPERIENCE

The University of Texas at Austin — Bullard Postdoctoral Research Fellow, School of Information*Sep 2024 – Present*
Postdoctoral researcher at the Information eXperience (IX) Lab, developing real-time data pipelines, behavioral analytics systems, and workflows for multimodal physiological and eye-tracking datasets.

- Designed and implemented **end-to-end data collection, processing, and analysis pipelines** for large-scale eye-tracking and physiological datasets, enabling real-time measurement of attention and cognitive load. Applied mixed-effects models, ANOVA, non-parametric statistical methods, and time-series analysis to extract behavioral insights and support interdisciplinary research initiatives. (*CHIIR 2025-2026, ETRA 2025-2026*)
- Analyzed large-scale eye-tracking datasets** to identify gaze-based indicators of misinformation susceptibility and investigate how document relevance and confirmation bias influence cognitive load, generating insights for information access, health literacy, and human behavior research. (*ETRA 2026, CHIIR 2025*)
- Developed real-time physiological signal processing pipelines** using Savitzky–Golay filtering, normalization, and feature engineering techniques to transform raw pupillometric sensor streams into live cognitive load estimates, outperforming existing offline approaches. (*JEMR 2025*)
- Designed and deployed real-time visual attention **classification pipelines** from eye-tracking data, generating structured behavioral metrics for downstream analytics, experimentation, and decision-making. (*ETRA 2025*)
- Collaborated** with researchers, domain experts, and stakeholders to define data requirements, design experimental protocols, and communicate analytical findings through publications, presentations, and technical reports.
- Ongoing:* Coordinating **multimodal behavioral data collection** and **developing analytical infrastructure** for ongoing studies of user interactions and learning outcomes in LLM-assisted learning environments.

Old Dominion University — Graduate Research & Teaching Assistant, Dept. of CS*Jan 2019 – Aug 2024*
Architected **RAEMAP**, a **real-time eye movement data streaming pipeline** ingesting raw gaze data via WebSocket, extracting 10+ behavioral features (fixations, saccades, entropy, cognitive load indices), and serving live analytical outputs to downstream applications and researchers. (*Ph.D. dissertation 2024 [Lee Entsminger Outstanding Dissertation Award]; ETRA 2022 [Best Doctoral Symposium]*)

- Redesigned and optimized pupillometric signal processing pipelines, enabling the **first real-time cognitive load estimation** framework from physiological sensor streams through advanced filtering, normalization, and feature extraction techniques. (*KES 2022*)
- **Built end-to-end machine learning pipelines** (tree-based classifiers, GCN), on structured biosensor datasets, including feature engineering, model training, validation, and deployment workflows, achieving **91%** prediction accuracy for ADHD classification. (*IEEE IRI 2019; Book Chapter 2019, MERCon 2019*)
- Developed **computer vision** and **data extraction pipelines** using YOLO and Mask R-CNN to automate area-of-interest detection in video stimuli, transforming unstructured multimedia data into structured datasets for large-scale behavioral analysis. (*IEEE IRI 2020; IJMDEM 2021*)
- Conducted large-scale behavioral data analyses to identify reading and information-processing patterns from eye-tracking datasets, earning **two Best Poster Awards** for research impact and analytical innovation. (*JCDL 2020, 2021*)
- Co-authored a highly cited (**379+**) systematic review synthesizing eye-tracking and pupillometric measures across clinical and public health domains, supporting data-driven decision making. (*Frontiers CS 2022*)
- Delivered undergraduate and graduate instruction in Python programming, database systems, software development, and computer architecture while mentoring student research and technical projects.

Los Alamos National Laboratory — *Summer Graduate Research Intern* (twice) *Jun–Aug 2021, Jun–Aug 2022*

- Designed and deployed an **NLP-driven document processing pipeline** (Flask + GROBID) to extract and structure metadata, section content, and references from **large scientific PDF corpora**. Automated document review workflows, eliminated manual data entry, and enabled scalable analysis across thousands of research documents in a national laboratory environment. (*LANL Tech. Report, 2022*)

UStocktrade (Pvt) Ltd, Sri Lanka — *Software Engineering Intern* *Jun – Dec 2017*

- Architected and implemented **MSstack**, a production-ready microservices framework with abstracted lifecycle management and data partitioning demonstrating scalable platform engineering and distributed data systems design. (*ICTer 2018*)

SELECTED PUBLICATIONS [[Google Scholar](#), 665+ citations, h-index: 11]

- [Jayawardena](#), Zhang, Gwizdka. "Phase-Dependent Scanpath Dynamics in Map-Based Tasks." *ETRA 2026*.
- [Jayawardena](#), Shi, Gwizdka. "Effects of Working Memory Capacity and Search Task Complexity on Cognitive Load." *CHIIR 2026*.
- Shi, [Jayawardena](#), Gwizdka. "Pupillometric Analysis of Cognitive Load in Relation to Relevance and Confirmation Bias." *CHIIR 2025*.
- [Jayawardena](#), et al. "Measuring Mental Effort in Real Time Using Pupillometry." *Journal of Eye Movement Research 18(6), JEMR 2025*.
- [Jayawardena](#), et al. "A Real-Time Approach to Capture Ambient and Focal Attention in Visual Search." *ETRA 2025*.
- [Jayawardena](#), et al. "Toward a Real-Time Index of Pupillary Activity as an Indicator of Cognitive Load." *KES 2022 – Procedia CS*.
- Mahanama, ..., [Jayawardena](#), et al. "Eye Movement and Pupil Measures: A Review." *Frontiers in Computer Science, 2022*. [**379 citations**]
- [Jayawardena](#), et al. "PDF Server: Extracting Metadata from Scientific PDF Documents." No. LA-UR-22-29460. Los Alamos National Laboratory (LANL), 2022.
- Mahanama, [Jayawardena](#), et al. "Analyzing Unconstrained Reading Patterns of Digital Documents." *JCDL 2021*. [**Best Poster**]
- De Silva, ..., [Jayawardena](#), et al. "A Rule-Based System for ADHD Identification." *MERCon 2019*. [**Best Paper**]

LEADERSHIP, SERVICE & AWARDS

Awards & Honors

- [**2024**] Lee Entsminger Outstanding Dissertation Award - *Old Dominion University, Norfolk, VA*
- [**2022**] Best Doctoral Symposium Presentation Award - *ETRA 2022*
- [**2020-21**] Best Poster Award - *ACM/IEEE JCDL 2021, 2020*
- [**2019-21**] Dominion Scholar; CS Outstanding Researcher Award; Levinstein Scholarship - *Old Dominion University*
- [**2019**] Best Paper Award - *IEEE MERCon 2019*

Professional Service

- [**2018-2023**] **Leadership** - President of ACM Student Chapter-Women, Old Dominion University 2022-23; President of ACM Student Chapter, University of Moratuwa 2018
- [**Ongoing**] **Reviewer/Editor** - ACM CHI, ETRA, CHIIR, OzCHI, IUI, JCDL, CIKM, AH, Frontiers in Psychology (Cognition), Acta Psychologica, Frontiers in Virtual Reality, Interacting with Computers - Oxford Academic
Reviewer experience spans human factors, cognitive science, information retrieval, and clinical/behavioral research venues.
- [**2019-present**] **Undergraduate/Graduate Research Mentor** - Supervised students on eye-tracking & gaze analysis projects at Computer Science Department, Old Dominion University; IX Lab, School of Information, The University of Texas at Austin; and NSF Research Experiences for Undergraduates (REU 2023)